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Review Article

Integrative characterization and DNA barcoding of *Fusarium* isolates from diverse agroecosystems

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Abstract

Fusarium comprises diverse filamentous fungi causing economic losses through pathogenicity and mycotoxin production. Identifying *Fusarium* species is crucial for understanding ecology, pathogenicity, and agricultural management. This review discusses advances in characterizing *Fusarium* from agroecosystems, including sampling, cultural, morphological, biochemical evaluations, pathogenicity research, and molecular analysis. Collections from infected plants and rhizospheric soils reveal extensive phenotypic and genetic diversity. Traditional assessments are limited by phenotypic variability under environmental conditions. Biochemical tests enhance differentiation but often lack species-level identification. The ITS sequence is the gold standard for species identification using molecular techniques. Phylogenetic analyses of ITS and loci (e.g., TEF-1α, RPB2) reveal clusters and evolutionary history. This review emphasizes the need for a local barcode reference library combining morphological and sequence data for rapid identification in outbreaks. Pathogenicity evaluation on crops associates genotype with phenotype and informs host–pathogen interaction studies and breeding. Coordinated *Fusarium* monitoring and molecular surveillance with data sharing are critical for designing location-specific disease control. This integrated approach serves as a framework for understanding *Fusarium* biodiversity and research on plant pathogenic fungi.

Keywords: Agricultural management, DNA barcoding, *Fusarium* spp., ITS sequencing, Morpho-molecular characterization, Phylogenetic analysis, Plant-pathogenic fungi.

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1. Introduction

Fusarium species are among the most important plant pathogenic fungi, causing diseases in major crops worldwide.[1,2] These diseases include wilt, root rot, crown rot, and head blight, causing yield losses and poor grain quality in cereals, legumes, vegetables, and fruits. *Fusarium*-associated diseases are worsened by mycotoxins like fumonisins, trichothecenes, and zearalenone, which risk human and livestock health through contaminated food.[3,4] These impacts necessitate expanding our knowledge of *Fusarium* diversity and pathogenicity for developing management options. However, identifying *Fusarium* spp. remains challenging due to species richness and morphological plasticity. Classical identification using cultural and morphological characters is often inadequate for species delimitation, due to overlapping traits and environmental effects on phenotypic expression.[5]

The limitations of taxonomy based on morphology have been highlighted by the presence of cryptic species complexes within *Fusarium*, even when these cryptic species exhibit minimal morphological variation. These hidden groups of taxa have been identified in several significant pathogen groups, such as the *Fusarium oxysporum* species complex (FOSC) and the *F. solani* species complex (FSSC), emphasizing that traditional methods fail to capture the complete diversity within this genus.[6,7] Additionally, interspecies hybrids, horizontal gene transfers, and anastomoses are intricately linked to species boundaries and often complicate the process of defining species solely based on their phenotypes. Consequently, botanical pathologists and mycologists have recognized the necessity of incorporating molecular data into taxonomic

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practices to achieve accurate, reproducible, and universally comprehensible species identifications.[8]

Owing to the complex taxonomy involved, integrative approaches that combine phenotypic data from traditional methods, such as classical cultural and morphological characterization, with biochemical tests and molecular tools have been adopted to accurately identify *Fusarium spp*.[9] Molecular markers, like the internal transcribed spacer (ITS) region of ribosomal DNA and genes encoding proteins such as translation elongation factor 1 alpha (TEF 1α), RNA polymerase II subunits (RPB1 and RPB2), or β-tubulin, are frequently used to distinguish closely related species or to gain a deeper understanding of phylogenetic relationships.[10,11] The advancement of multi-locus sequence typing (MLST) and DNA barcoding schemes has significantly improved diagnostic precision, enabling researchers to identify and differentiate pathogenic lineages that are not otherwise distinguishable by their morphology.[12]

Despite inconsistent protocols for characterizing *Fusarium* across laboratories and regions, the lack of comprehensive barcode reference libraries with morphological, biochemical, and sequence data remains a major obstacle for comparing studies.[13] This impedes pathogen monitoring, strain identification, and resistance-breeding programs. Systematic reviews are needed to gather existing methods and suggest unified characterization strategies for widespread adoption.

This review summarizes approaches for describing *Fusarium* isolates from agroecosystems. We explored morphological methods, biochemical tests, and molecular techniques, particularly DNA barcoding, for species identification. We emphasized regional barcode libraries connecting genetic signatures with phenotypic traits for identification.(Figure 1)

Isolating and identifying *Fusarium* species involves sampling, isolation, and morphological description. Phenotypic data includes biochemical tests, while molecular characterization involves DNA extraction, PCR, and phylogenetic analysis. Data are compiled for diagnostics.

2. Systematic Collection and Preservation

The systematic gathering and preservation of isolates from field samples are fundamental to understanding *Fusarium* species diversity in agricultural ecosystems.[14,15] Effective sampling requires protocols focusing on *Fusarium* reservoirs.[15] Infected plant materials should be collected aseptically from lesion edges. Soil cores from symptomatic plants and rhizosphere (2–4 mm soil adhering to roots) contain dense *Fusarium* populations. Geography and crop diversity guide sampling across varied agroecosystems with different climates, soils, and practices. Sampling should cover diverse zones and crops to determine host specificity and reveal regional pathogen ecology. Surface sterilization and selective media are essential for *Fusarium* isolation. Plant tissues are washed with ethanol or sodium hypochlorite before plating on PDA, CLA, or *Fusarium* selective media. Soil samples are plated or baited to stimulate *Fusarium* growth.[16]

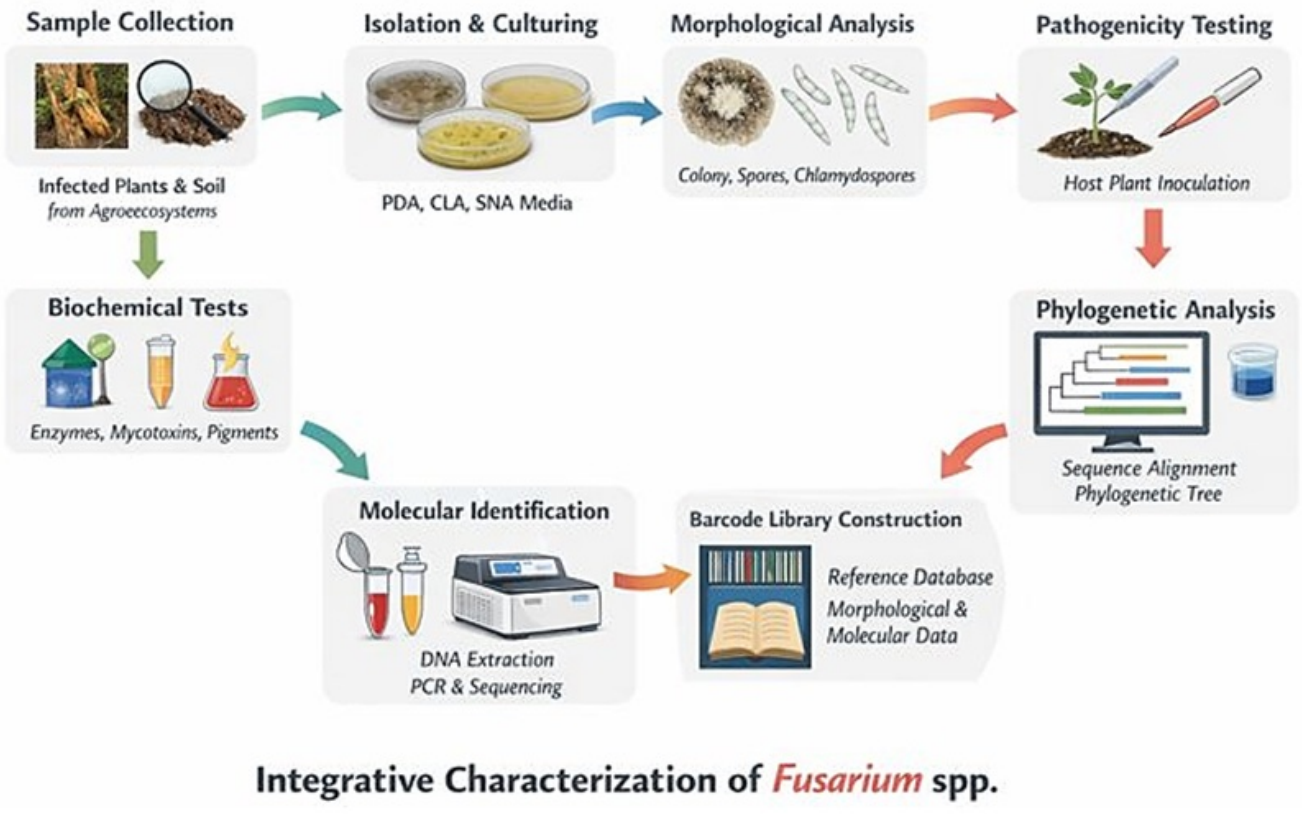


Figure 1: *Workflow for the integrated characterization of Fusarium spp.*

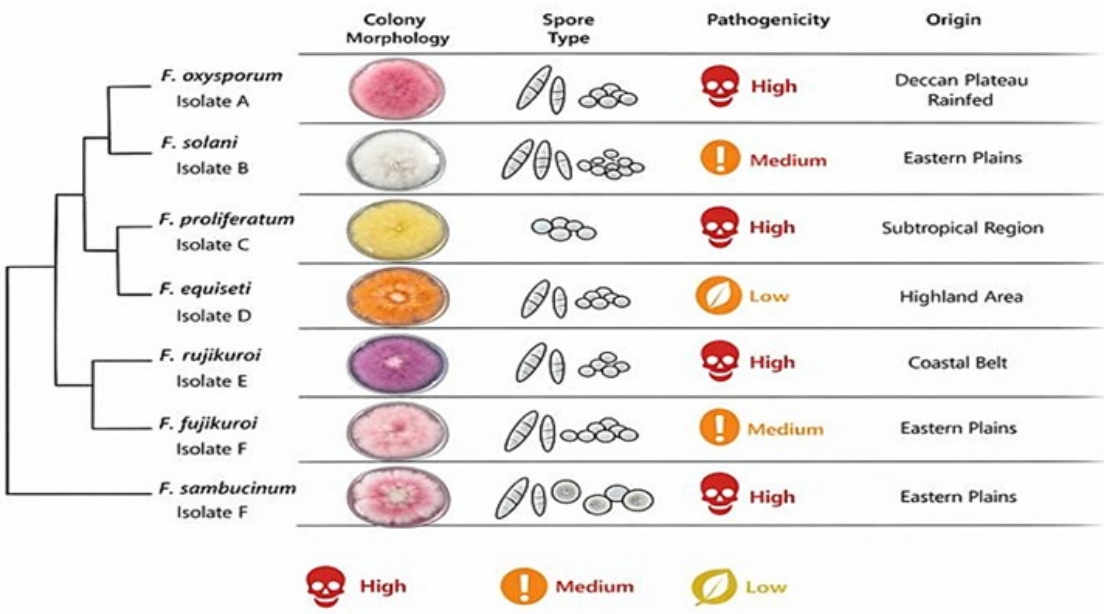


Figure 2: *Analysis of Fusarium phylogeny shows connections between isolates through ITS and TEF-1α sequences. Colony morphology, spores, pathogenicity, and origin are described. Pathogenicity levels use red skull (high), orange exclamation (medium), yellow leaf (low), showing genetic groups' traits*

Table 1. *Characterization and DNA barcoding of fusarium isolates*

Step	Procedure / Method	Purpose / Notes	References
Plant Tissue Sampling	Collect symptomatic tissues from lesion margins using sterile tools; store in moist, cool conditions	Ensures high pathogen load and recovery of viable Fusarium spp.	17, 18
Soil/Rhizosphere Sampling	Collect rhizospheric soil at 5–15 cm depth; use composite sampling near infected roots	Targets soil-residing Fusarium propagules and increases detection sensitivity	18, 19
Surface Sterilization	Treat samples with 70% ethanol or 1–1.5% sodium hypochlorite for 1–2 min, followed by sterile water rinse	Removes surface contaminants while preserving internal pathogens	20
Isolation Media	Plate onto PDA, CLA, or Fusarium Selective Medium (FSM); incubate at 25–28°C	Enhances selective recovery of Fusarium colonies	17,21
Morphological Confirmation	Observe macroconidia, microconidia, chlamydospore structures using compound microscopy	Confirms genus-level identity prior to molecular testing	22
Cryopreservation	Store culture in cryovials with 15–25% glycerol at −80°C or in liquid nitrogen	Long-term storage with minimal genetic drift	23
Mineral Oil Storage	Overlay cultures on slants with sterile mineral oil and store at room temperature or 4–10°C	Low-cost medium-term storage alternative	24
Lyophilization	Freeze-dry fungal cultures and seal under vacuum; store in sterile ampoules at 4°C	Best for long-term preservation, though rehydration protocols must be optimized	25
Metadata Documentation	Record host species, geographic coordinates, sample type, climate data, soil type, collection date, and cultural traits	Essential for future traceability, ecological studies, and comparative analysis	17,26

Preserving isolates maintains viability and stability for future studies. Cryopreservation in liquid nitrogen or at −80°C enables long-term storage, reducing metabolic activity and genetic changes, but needs special equipment. Mineral oil preservation involves inoculating mycelia into agar plugs with oil overlay at moderate temperatures. This method keeps *Fusarium* viable for years but requires monitoring. Lyophilization preserves cultures through vacuum dehydration at low temperature for decades if sealed. Method selection depends on infrastructure and use.

3. Cultural and Morphological Characterization

Culture and morphology-based characterization remains fundamental for *Fusarium* taxonomy. Isolates are cultured from infected plant tissues on nutrient media. PDA promotes mycelium growth and pigmentation between *Fusarium* spp.[27] Colony morphology varies in growth rate, texture, and pigmentation. CLA and SNA stimulate reproductive structures, with CLA triggering sporulation and SNA showing conidial morphology.[28] Macroconidia characteristics, including length, shape, and septation, distinguish species. Microconidia shape and formation patterns aid identification.[29] Chlamydospores help identify certain species. Cultural characterization has limitations due to phenotypic plasticity and trait overlap. Combining morphological data with molecular markers provides robust taxonomy. Morphological profiles validate molecular classifications and show phenotypic links to phylogenetic groups.

Table 2. *Morphological and cultural characteristics of common fusarium species*

Species	Colony Morphology (on PDA/CLA)	Macroconidia	Microconidia	Chlamydospores	References
F. oxysporum	White to purple; cottony/floccose; slow growth	Slender, 3–5 septa, slightly curved	Abundant, oval to ellipsoidal	Abundant, terminal/intercalary	17
F. solani	Fast-growing; white with bluish/green center	Broad, 3–5 septa, blunt ends	Oval or kidney-shaped, formed in chains	Present, terminal/intercalary	28,29
F. proliferatum	Pink to violet, dense aerial mycelium	Narrow, 3–5 septa, curved	Abundant, oval/reniform	Rare or absent	17
F. verticillioides	Cottony white to lilac, flat colonies	Narrow, slender, 3–5 septa, pointed ends	Oval, produced in long chains	Absent or very rare	30,31
F. equiseti	Orange-brown center, radial growth pattern	Thick-walled, curved, 5–7 septa	Sparse, oval	Abundant, globose	32
F. chlamydosporum	White to pale pink, rough mycelium	Small, slightly curved, 3–5 septa	Sparse, ovoid	Very abundant, diagnostic trait	33

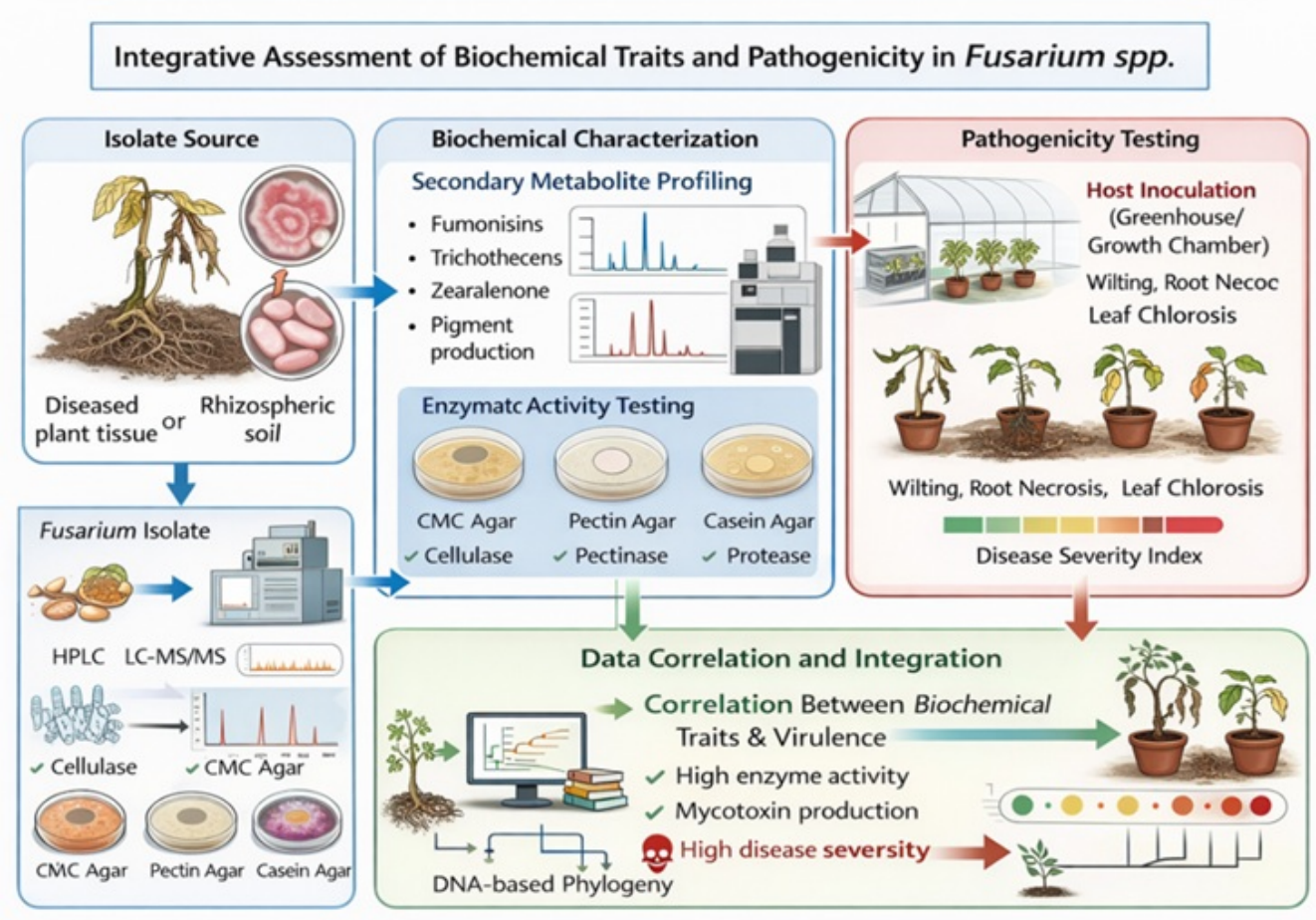


Figure 3: *Integrated biochemical and pathogenicity testing of Fusarium requires bacterial isolation. HPLC/LC-MS/MS screens metabolites and enzymatic activities. Host inoculation evaluates disease symptoms. Data integration links biochemical traits with virulence for diagnostics*

4. Biochemical and Pathogenicity Assessments

Secondary metabolite profiling distinguishes species based on metabolite production and pathogenic potential. *Fusarium* spp. produce mycotoxins (fumonisins, trichothecenes, zearalenone, and enniatins), which are phytotoxic and hazardous through food contamination. These metabolite profiles differ among species, serving as chemotaxonomic markers. Analytical methods (HPLC, GC MS, and LC MS/MS) detect these compounds in cultures or infected plants.[35,36] Specific secondary metabolites help distinguish related taxa that morphology cannot differentiate. Pigmentation helps help delineate *Fusarium* groups, with red, violet, and orange pigments being partly clade-specific. *Fusarium* pathogens secrete cell wall-degrading enzymes like cellulases and pectinases to break down host structures.[37] Substrate assays measure enzymatic activity, with higher activity indicating increased aggressiveness. While enzyme profiles don't define species, they help characterize virulence patterns.

Host-specific pathogenicity tests evaluate disease through seedling inoculation and symptoms. Virulence status differentiates pathogenic from opportunistic strains. Disease severity links to genetic differences, producing indices that correlate to molecular markers.[38] Biochemical profiling

connects phenotypic features to virulence. Isolates with higher enzyme activity cause severe symptoms. These correlations help select markers for risk analysis. Linking metabolites to enzyme activities and pathogenicity improves understanding of *Fusarium* control.

5. Molecular Identification and Phylogenetics

Molecular biology enables precise identification of *Fusarium* species, overcoming morphological taxonomy limitations. The process involves isolating genomic DNA from fungal cultures using mechanical disruption and chemical lysis. DNA purity is verified through spectrophotometric analysis and electrophoresis. Specific primers targeting conserved regions amplify loci via PCR. The internal transcribed spacer (ITS) region serves as a barcode marker in fungal systematics, showing interspecific variation. ITS1, 5.8S rRNA gene, and ITS2 enable species distinction. ITS sequencing allows database comparison, although this locus may lack sufficient resolution within species complexes (Table 3). Multiple locus sequence typing provides greater phylogenetic resolution. Protein-coding genes like translation elongation factor 1 alpha (TEF-1α), RNA polymerase II subunits (RPB1 and RPB2), and β-tubulin serve as markers.[40] TEF-1α shows strong discrimination between *Fusarium* species, particularly within *F.* oxysporum species complex (FOSC) or *F. solani* species complex (FSSC) (Table 2). Multiple loci enable dataset concatenation, improving phylogenetic interpretation.

6. DNA Barcoding and Construction of Reference Library

Molecular identification and phylogeny enable precise determination of *Fusarium* species through DNA isolation and PCR amplification using specific primers.[41] The internal transcribed spacer (ITS) region serves as a fungal barcode locus, although it may lack sufficient resolution among Fusarium species complexes. Multilocus sequence typing (MLST) uses protein-coding genes TEF-1α, RPB1, RPB2, and β-tubulin for better phylogenetic resolution (Table 2), with TEF-1α showing strong species discrimination.[2,42]

Phylogenetic trees use sequence alignments as terminal nodes, analysed through Maximum Likelihood, Bayesian Inference, and Neighbour Joining methods using MEGA, RAxML, and MrBayes.[43,44] Well-supported clades link to morphological traits and pathogenic behavior. Multilocus phylogenies reveal distinct lineages with unique epidemiological profiles,[45] enabling predictive identification for disease management.

7. Disease Management and Breeding Application

DNA barcoding of *Fusarium* isolates aids disease management and crop breeding. Using defined isolates enhances resistance tests in breeding programs. Conventional breeding uses limited pathogen isolates, but genetic variation suggests resistance may be lost with divergent pathogen compositions. Isolates characterized through morphological and molecular examination improve resistance screening.[51] DNA barcoding helps identify virulence genes for marker-assisted selection.[41] DNA barcodes enable early pathogen detection for disease management through molecular tools that identify infections when fungicide treatments are most effective.[51,52] Surveillance systems using barcoding can be implemented through extension services. Early detection enables intervention through seed treatment or altered planting dates. Pathogen characterization enables targeted control since *Fusarium* populations vary regionally.[1] Control methods may not work across species due to host differences.[25,54]

8. Challenges and Future Directions

Despite advances in integrative characterization of *Fusarium* species, a comprehensive global species identification system has limitations. A major issue is incomplete barcode coverage for *Fusarium* species in public repositories.[5] Many plant disease-causing microorganisms lack high-quality sequence data with verified strains and pathogenicity, especially those underrepresented in studies and minor crops. This gap weakens molecular identification and limits phylogenetic studies. Additionally, inconsistent protocols between laboratories for cultivation, morphological documentation, and DNA sequencing reduce reproducibility and hinder data integration.[55]

Table 3. *Common molecular markers used for the identification and phylogenetic analysis of fusarium spp*

Locus	Target Region	Common Primer Pairs	Primary Application	Reference
ITS	Internal Transcribed Spacer (ITS1–5.8S–ITS2)	ITS1: 5′-TCCGTAGGTGAACCTGCGG-3′ ITS4: 5′-TCCTCCGCTTATTGATATGC-3′	Universal fungal barcode; species-level identification	46
TEF-1α	Translation Elongation Factor 1-alpha	EF1: 5′-ATGGGTAAGGA(A/G)GACAAGAC-3′ EF2: 5′-GGA(G/A)GTACCAGT(G/C)ATCATGTT-3′	High-resolution species delimitation; MLST in complexes	47
RPB2	RNA Polymerase II subunit 2	fRPB2-5f: 5′-GAYGAYMGWGATCAYTTYGG-3′ fRPB2-7cr: 5′-CCCATRGCTTGYTTRCCCAT-3′	Phylogenetic resolution in deeper clades	48
β-tubulin	β-tubulin gene region	Bt2a: 5′-GGTAACCAAATCGGTGCTGCTTTC-3′ Bt2b: 5′-ACCCTCAGTGTAGTGACCCTTGGC-3′	Species-level differentiation; MLST	49
RPB1	RNA Polymerase II subunit 1	RPB1-Af: 5′-CAYCCWGGYTTYATCAAGAA-3′ RPB1-Cr: 5′-CCNGCDATNTCRTTRTCCATRTA-3′	Complementary to TEF and RPB2 in phylogenetic studies	50

Connecting traditional morphology-based identification with new methods like high-throughput sequencing (HTS) and metagenomics for non-culturable *Fusarium* species detection in environmental samples is challenging.[56,57] These methods complement culture-dependent techniques but need well-curated reference databases and bioinformatics pipelines. Integrating classical taxonomy with HTS data is crucial for uncovering cryptic diversity and monitoring

pathogen emergence. To address these issues, developing cooperative research networks and public barcode libraries is essential.[57,58]

9. Discussion

The integrative *Fusarium* characterization represents a breakthrough for fungal taxonomy, plant pathology, and agricultural biosecurity. Precise diagnostics cannot rely solely on morphology due to phenotypic plasticity among cryptic species.[19,59] Biochemical, pathogenicity, and molecular data, especially ITS and TEF-1α sequencing, enable definitive identification.

Our synthesis shows variation among *Fusarium* populations across agroecological zones. Host selection and environmental pressures that structure diversity impact disease control.[2] Pathogenic isolates vary in virulence and toxin production, requiring connection between molecular identification and phenotypic traits. Phylogenetic analyses provide insights into evolutionary relationships, helping predict threats 8. Building DNA barcode libraries facilitates specimen identification. Challenges remain in achieving consensus taxonomy given varied methodologies. High-throughput sequencing and metagenomics will advance understanding of *Fusarium* biodiversity.[60,61] Harmonized data sharing and collaboration are needed to realize integrative characterization's potential for improved disease monitoring and crop protection.

10. Conclusion

Fusarium species identification and characterization are necessary to study pathogenicity, distribution, and agricultural vulnerability. This study highlights deficiencies in morphological-based methods and the need for cultural, biochemical, pathogenicity, and molecular characteristics. ITS and TEF-1α sequences with phylogenetic analysis could serve as a consistent method for identifying closely related Fusarium species. Local DNA barcode libraries linked to phenotypic and genotypic information accelerate *Fusarium* diagnosis without full multilocus analysis. These approaches are important for breeding programs, quarantine restrictions, and developing regional disease control strategies. Limitations include non-standardized protocols, incomplete barcode coverage, and limited high-throughput sequencing. These must be addressed through research networks, open-access platforms, and collaborations for advancement.

11. Source of Funding

None.

12. Conflict of Interest

None.

13. Ethics Approval

This research did not engage with human subjects nor animals, nor any clinical/personal/facility data. All the fungi were isolated from environmental and agricultural sources only. Therefore, ethical approval was not required for this study. All experiments were performed according to standard laboratory practices and institutional biosafety guidelines for microbial handling.

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